

Amir Qureshi

Embedded Robotics Engineer

Karachi, Pakistan | amir.qureshi.demo@example.com | +92 300 111 2201
github.com/amirqureshi-robotics | amir-embedded.dev

Professional Summary

Embedded robotics engineer with hands-on Arduino and ESP32 experience building line-following robots, sensor-driven automation systems, and control logic in C++. Comfortable with rapid prototyping, field testing, and debugging.

Technical Skills

Arduino, ESP32, C++, Motor and sensor integration, Embedded debugging, Prototype testing, React, Node.js

Experience Highlights

- Built and tuned two line-following robots for university competitions using C++ control logic and IR sensor calibration.
- Integrated ESP32 motor drivers, ultrasonic sensors, and telemetry dashboards for a warehouse robotics prototype.
- Developed a lightweight Node.js API and React dashboard for testing robot runs and logging sensor behaviour.

Selected Projects

- Autonomous cart prototype using ESP32, PID-style tuning, and motor encoder feedback.
- Workshop management web app for hardware issue tracking and test result logging.

Education

Bachelor-level Computer Science or Engineering background with project-based robotics, embedded, software, or automation exposure through coursework, competitions, freelance work, or internships.